A product counterexample to the simplex maximum for projection-body volume

Rewritten for readability from the OpenAI manuscript of the same title

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Abstract

The product of two ten-dimensional simplices has larger normalized projection-body volume than a twenty-dimensional simplex. This gives a counterexample to Brannen's proposed simplex maximum.

1 The result and how it is proved

1.1 Setting and definitions

Throughout, $d \geq 2$. A *convex body* in $\mathbb{R}^d$ is a compact convex set with nonempty interior. We write $|K|$ for its d-dimensional volume, $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$ for the Euclidean inner product and norm, S^{d-1} for the unit sphere, B_2^j for the Euclidean unit ball of $\mathbb{R}^j$, and $\kappa_j = |B_2^j|$. For $u \in S^{d-1}$, $\mathrm{proj}_{u^\perp}$ is the orthogonal projection onto the hyperplane $u^\perp$. The *support function* of a compact convex set C is $h_C(x) = \max_{y \in C} \langle x, y \rangle$. The *Minkowski sum* is $C + D = \{c + d : c \in C,\ d \in D\}$, and $[p, q]$ denotes the segment from p to q.

Projection body. For a convex body $K \subset \mathbb{R}^d$, the projection body ΠK (also $\Pi_d K$) is the compact convex set whose support function is

$$h_{\Pi K}(u) = \mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} K) \quad (u \in S^{d-1}), \qquad h_{\Pi K}(tu) = t\, h_{\Pi K}(u) \quad (t \geq 0).$$

This is the definition throughout. A compact convex set is determined by its support function (fact (S4) in Section 2), so ΠK is unique if it exists. Existence is proved for polytopes in Lemma 2, with an explicit formula, and for arbitrary convex bodies in Step 6 of the proof of Lemma 4. The main theorem and the corollary involve polytopes only.

The functional and the simplex constant.

$$R_d(K) = \frac{|\Pi K|}{|K|^{d-1}}, \qquad c_d = \frac{(d+1)d^d}{d!}. \tag{1}$$

Proposition 5 shows that $R_d(T) = c_d$ for every d-simplex T. Brannen conjectured in 1996 that simplices maximize R_d among all convex bodies [1]. In view of the simplex value, this is the proposed bound

$$|\Pi K| \leq c_d |K|^{d-1} \qquad \text{for every convex body } K \subset \mathbb{R}^d. \tag{2}$$

1.2 Main results

Theorem 1. *Let $T_{10} = \operatorname{conv}(0, e_1, \dots, e_{10}) \subset \mathbb{R}^{10}$, where the e_i are the coordinate unit vectors, and let $K = T_{10} \times T_{10} \subset \mathbb{R}^{20}$. Then*

$$\frac{R_{20}(K)}{c_{20}} = \frac{121\binom{20}{10}}{21 \cdot 2^{20}} = \frac{22\,355\,476}{22\,020\,096} > 1.$$

Thus K violates (2) in dimension twenty.

The second result, Corollary 6 in Section 4, says that the excess over the simplex value can be made exponential in the dimension: there are $\lambda > 1$ and N such that for every $n \geq N$ some product of simplices $K_n \subset \mathbb{R}^n$ has $R_n(K_n) \geq \lambda^n c_n$.

1.3 Context (reported from the source; not used in any proof)

Everything in this subsection is taken from the source's introduction. It rests on the cited papers, which this document does not check, and no proof below uses it.

Petty established the affine covariance of the projection-body operator [6]; see also [5, Equation (1)]. The functional R_d involves the volume of ΠK itself, not of its polar; the classical projection inequalities involving the polar concern a different functional. For $d \geq 3$, Petty's separate minimum problem asks whether ellipsoids are the only minimizers of the same functional R_d; see [7, Section 1] for these distinctions.

In dimension three, Chen, Feng, Li, Xi, and Xu settle the minimum problem and also prove $R_3(K) \leq 18 = c_3$ for all three-dimensional convex bodies, with tetrahedra attaining equality [2, Theorems 1.1 and 1.2]. Feng, Hu, Liu, and Xu give counterexamples to (2) in every dimension $d \geq 9$, using polytopes with at most $d + 2$ facets [3, Theorem 1.3]; their construction uses perturbed projections of a cube and Minkowski's existence theorem [3, Section 4]. Taken at face value, that cited theorem already shows that (2) fails in dimension twenty. So the failure of (2) is not new with the present paper. The source says so: "Their result already gives a negative answer to the unrestricted-dimensional simplex-maximum question." It describes its own contribution as "a direct Cartesian-product construction and its exact value: two ten-dimensional simplices suffice". Theorem 1 is proved below without using any cited result. The optimal upper constant and its maximizing bodies remain separate questions.

For centrally symmetric bodies, Henk gives the lower bound $2^d(9/8)^{\lfloor d/3 \rfloor}$ for the optimal upper constant [4]. This sentence is the source's. In its context, "upper constant" refers to upper bounds for R_d, here over centrally symmetric bodies. The source quotes it only as background on how such constants grow with d, and it draws no comparison with Corollary 6. That corollary measures growth *relative to the simplex value* c_n only.

1.4 Proof at a glance

Key idea. For polytopes, R is multiplicative under Cartesian products:

$$R_{r+s}(A \times B) = R_r(A)\, R_s(B) \qquad (r, s \geq 2).$$

The simplex constants are not: $c_{10}^2 > c_{20}$. Hence $R_{20}(T_{10} \times T_{10}) = c_{10}^2 > c_{20}$. In general, call C a *universal upper constant in dimension d* if $R_d(K) \leq C$ for all convex bodies $K \subset \mathbb{R}^d$. Since $T_r \times T_s$ is a convex body in $\mathbb{R}^{r+s}$ with $R_{r+s} = c_r c_s$, every such constant in dimension $r + s$ is at least $c_r c_s$. The conjecture would need $c_r c_s \leq c_{r+s}$.

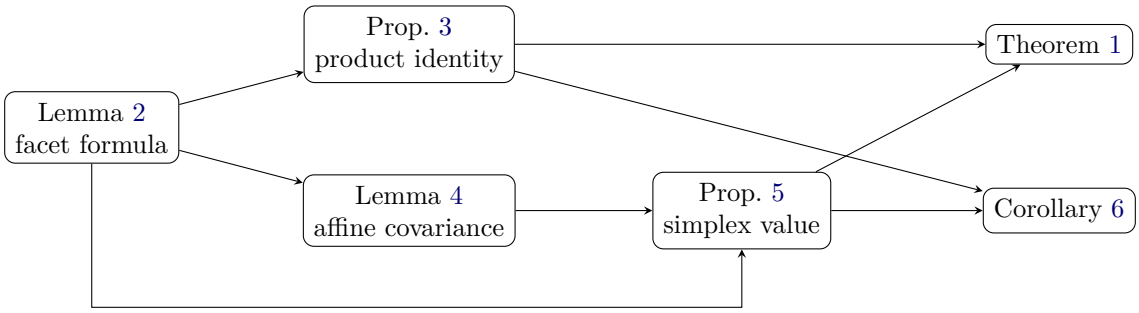
An informal heuristic, not used in any proof, explains why this fails. Stirling's formula $d! \approx \sqrt{2\pi d}\,(d/e)^d$ gives $c_d \approx (d+1)e^d/\sqrt{2\pi d}$, hence

$$\frac{c_r c_s}{c_{r+s}} \approx \frac{(r+1)(s+1)}{r+s+1}\sqrt{\frac{r+s}{2\pi rs}}.$$

The exponential factors have cancelled. For $r = s$ the right side is $(r+1)^2/((2r+1)\sqrt{\pi r})$, which grows like $\sqrt{r}/(2\sqrt{\pi})$. At $r = 10$ it is about 1.03; the exact value, from Theorem 1, is about 1.015.

Steps.

1. *Facet formula (Lemma 2).* For a polytope P, ΠP is a sum of segments, one per facet, each spanned by the facet's unit normal scaled by its area. Everything else is computed from this.
2. *Product identity (Proposition 3).* The facets of $A \times B$ are $F \times B$ and $A \times G$. Step 1 then gives $\Pi(A \times B) = (|B|\Pi A) \times (|A|\Pi B)$, and taking volumes gives multiplicativity.
3. *Affine covariance (Lemma 4).* $R_d(LP) = R_d(P)$ for invertible linear L. It is used only to pass from the standard simplex to all simplices.
4. *Simplex value (Proposition 5).* By Step 1, ΠT_d is a scaled translate of "unit cube plus one segment", whose volume is $d+1$. This gives $R_d(T_d) = c_d$.
5. *Arithmetic.* Theorem 1 is the exact evaluation of c_{10}^2/c_{20}. Corollary 6 iterates the product identity and uses the bound $c_n < (n+1)(11/4)^n$.



Two remarks for a reader short of time.

- Theorem 1 and Corollary 6 use only the *standard* simplex part of Proposition 5, which does not need Lemma 4. Lemma 4, in its polytope case, is needed only for the sentence "every simplex has value c_d", which identifies c_d as the simplex value in Brannen's conjecture.
- The second half of Lemma 4 (arbitrary convex bodies, Steps 5–8) is used nowhere else. It is kept because the source includes it and because it shows that ΠK exists for every convex body.

1.5 Notation changes relative to the source

Four symbols are renamed to avoid clashes, with no change in the mathematics. In the proof of Lemma 4, the outer radius R becomes ρ (the letter R_d is the functional), the factors λ_m become μ_m (λ is the constant of Corollary 6), and the functions F, f_m become φ, φ_m (F is a facet). In the proof of Corollary 6, the constant A becomes α (A is a polytope in Proposition 3).

2 Facet vectors and Cartesian products

2.1 Four facts about support functions

Let C, D be compact convex sets. Each of (S1)–(S3) is immediate from $h_C(x) = \max_{y \in C} \langle x, y \rangle$.

(S1) $h_{C+D} = h_C + h_D$, and $h_{tC} = t\, h_C$ for $t \geq 0$. The maximum of $\langle x, c + d \rangle$ over independent $c \in C$, $d \in D$ splits into two maxima.

(S2) For $C \subset \mathbb{R}^r$, $D \subset \mathbb{R}^s$ and $(u, v) \in \mathbb{R}^r \times \mathbb{R}^s$: $h_{C \times D}(u, v) = h_C(u) + h_D(v)$, for the same reason. For a linear map M: $h_{MC}(u) = h_C(M^t u)$, because $\langle u, Mc \rangle = \langle M^t u, c \rangle$.

(S3) $h_{[-v/2,\, v/2]}(u) = \frac{1}{2} |\langle v, u \rangle|$, since a linear function on a segment is maximal at an endpoint. Also $h_{tB_2^d}(u) = t\|u\|$.

(S4) If $h_C \leq h_D$ on $\mathbb{R}^d$, then $C \subset D$. In particular $h_C = h_D$ implies $C = D$.

Proof of (S4). Suppose $x \in C \setminus D$. Let $y \in D$ be the point of D nearest to x, so $x \neq y$. For $z \in D$ and $t \in [0, 1]$ the point $y + t(z - y)$ lies in D, so $g(t) = \|x - y - t(z - y)\|^2$ is minimal at $t = 0$. Hence $g'(0) = -2\langle x - y, z - y \rangle \geq 0$. Thus $\langle x - y, z \rangle \leq \langle x - y, y \rangle$ for all $z \in D$, that is, $h_D(x - y) = \langle x - y, y \rangle$. But $\langle x - y, x \rangle = \langle x - y, y \rangle + \|x - y\|^2 > h_D(x - y)$, while $x \in C$ gives $\langle x - y, x \rangle \leq h_C(x - y) \leq h_D(x - y)$. This is a contradiction. □

Support functions are positively homogeneous ($h_C(tx) = t\, h_C(x)$ for $t \geq 0$). So an identity or inequality between support functions need only be checked on S^{d-1}.

2.2 The facet formula

We use the following standard facts about a full-dimensional convex polytope $P \subset \mathbb{R}^d$. It has finitely many faces, and its *facets* are the faces of dimension $d - 1$. For a facet F, write s_F for its $(d-1)$-dimensional volume, ν_F for its outward unit normal, and $H_F = \{x : \langle \nu_F, x \rangle = h_P(\nu_F)\}$ for its hyperplane. Then:

(P1) $P = \bigcap_F \{x : \langle \nu_F, x \rangle \leq h_P(\nu_F)\}$ and $F = P \cap H_F$.

(P2) The boundary of P is the union of the facets.

(P3) A point of a facet F outside the relative interior of F, and a point lying in two distinct facets, lies in a face of dimension at most $d - 2$.

We call $s_F \nu_F$ the *area-normal vector* of F.

The next lemma is the polytopal case of Cauchy's projection formula; see [7, Section 1]. The proof is included to fix the factor $1/2$, on which all later constants depend. The lemma is used in Proposition 3, Lemma 4 and Proposition 5.

Lemma 2. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a full-dimensional convex polytope. Then ΠP exists,*

$$h_{\Pi P}(u) = \frac{1}{2} \sum_F s_F |\langle \nu_F, u \rangle|, \qquad u \in \mathbb{R}^d, \tag{3}$$

and therefore

$$\Pi P = \sum_F \left[-\frac{s_F \nu_F}{2}, \frac{s_F \nu_F}{2} \right], \tag{4}$$

where both sums run over the facets F of P.

Proof. Plan. Fix a unit vector u and let $\pi = \mathrm{proj}_{u^\perp}$. We show that π scales areas on H_F by $|\langle \nu_F, u\rangle|$ (Step 1). We then show that almost every point of $\pi(P)$ is covered by the projections of exactly two facets (Step 2). Counting the covered area in two ways gives (3) for unit u (Step 3). The general case and (4) follow from (S1), (S3), (S4) (Step 4).

Step 1 (Jacobian). *For every facet F and measurable $S \subset H_F$,* $\mathrm{vol}_{d-1}(\pi(S)) = |\langle \nu_F, u\rangle| \,\mathrm{vol}_{d-1}(S)$. The restriction of π to H_F is affine with linear part $J\colon \nu_F^\perp \to u^\perp$, $Jv = v - \langle v, u\rangle u$. So the claim is $|\det J| = |\langle \nu_F, u\rangle|$; when this is 0, both sides vanish. Let $v_1, \dots, v_{d-1}$ be an orthonormal basis of $\nu_F^\perp$. Then $|\det J|$ is the $(d-1)$-volume of the parallelepiped spanned by $Jv_1, \dots, Jv_{d-1}$ in $u^\perp$. Since u is a unit vector orthogonal to all Jv_i, this equals the d-volume of the parallelepiped with the extra edge u:

$$|\det J| = |\det(Jv_1, \dots, Jv_{d-1}, u)| = |\det(v_1, \dots, v_{d-1}, u)| = |\langle u, \nu_F\rangle|\,|\det(v_1, \dots, v_{d-1}, \nu_F)| = |\langle u, \nu_F\rangle|.$$

The second equality adds multiples of the last column to the others. The third expands $u = \langle u, \nu_F\rangle \nu_F + \sum_i \langle u, v_i\rangle v_i$ in the last column, where the terms with a repeated column vanish. The last holds because $(v_1, \dots, v_{d-1}, \nu_F)$ is an orthonormal basis.

Step 2 (generic lines meet exactly two facets). For $y \in u^\perp$ let $\mathrm{mult}(y)$ be the number of facets F with $y \in \pi(F)$. Clearly $\mathrm{mult}(y) = 0$ for $y \notin \pi(P)$. *We claim* $\mathrm{mult}(y) = 2$ *for all* $y \in \pi(P) \setminus E$, *where* $E \subset u^\perp$ *has* $(d-1)$*-dimensional measure zero.*

The exceptional set. Let E be the union of $\pi(G)$ over all faces G of dimension at most $d-2$, together with $\pi(F)$ over all facets F with $\langle \nu_F, u\rangle = 0$. Each $\pi(G)$ lies in the image of an affine space of dimension at most $d-2$. If $\langle \nu_F, u\rangle = 0$, then H_F contains the direction u, so $\pi(H_F)$ has dimension $d-2$. Thus E is a finite union of subsets of affine subspaces of $u^\perp$ of dimension at most $d-2$, hence null.

The count. Let $y \in \pi(P) \setminus E$ and $\ell = y + \mathbb{R}u$; every point of ℓ projects to y. By compactness and convexity, $\ell \cap P$ is a nonempty segment $[p, q]$, possibly with $p = q$. Both endpoints lie on the boundary of P, because ℓ leaves P there.

(i) By (P2), p lies in a facet F_p. If p lay in a face G of dimension at most $d-2$, then $y = \pi(p) \in \pi(G) \subset E$, which is excluded. So by (P3), p is in the relative interior of F_p and in no other facet. Also $\langle \nu_{F_p}, u\rangle \neq 0$: otherwise $y = \pi(p) \in \pi(F_p) \subset E$. The same holds for q, with a facet F_q.

(ii) By (P1) and (i), every facet inequality other than that of F_p is strict at p. So in a small ball around p, P coincides with the half-space $\{\langle \nu_{F_p}, x\rangle \le h_P(\nu_{F_p})\}$. As $\langle \nu_{F_p}, u\rangle \neq 0$, the points $p + tu$ with $|t|$ small, $t \neq 0$ and $t\langle \nu_{F_p}, u\rangle < 0$ are interior points of P. Fix one of them, x_0. It lies in $\ell \cap P = [p, q]$ and $x_0 \neq p$, so $p \neq q$. Also $x_0 \neq q$, because q is a boundary point. Thus x_0 lies strictly between p and q.

(iii) If x_0 is interior to P and $z \in P$, every point of $[x_0, z]$ other than z is interior to P. (If a ball of radius ε about x_0 lies in P, convexity puts a ball of radius $(1-t)\varepsilon$ about $(1-t)x_0 + tz$ in P.) Apply this with $z = p$ and $z = q$. Since x_0 is strictly between p and q, the two half-open segments cover the open segment between p and q, which therefore lies in the interior of P. So ℓ meets the boundary of P only at p and q.

(iv) A facet F with $y \in \pi(F)$ contains a point of ℓ. This point is a boundary point of P, hence is p or q by (iii), so $F \in \{F_p, F_q\}$ by (i). Conversely $y \in \pi(F_p)$ and $y \in \pi(F_q)$. Finally $F_p \neq F_q$. The function $t \mapsto \langle \nu_{F_p}, y + tu\rangle$ has slope $\langle \nu_{F_p}, u\rangle \neq 0$, so H_{F_p} meets ℓ in exactly one point, namely p. If $F_q = F_p$, then $q \in H_{F_p} \cap \ell = \{p\}$, contradicting $p \neq q$.

Hence $\mathrm{mult}(y) = 2$.

Remark. The source's exceptional set also contains the relative boundary of $\pi(P)$. Discarding it is harmless but not needed. By (ii), for $y \in \pi(P) \setminus E$ the line ℓ passes through an interior point of P, so y is an interior point of $\pi(P)$ in $u^\perp$.

Step 3 (double counting). Integrating the indicator functions of the sets $\pi(F)$ over $u^\perp$ and using Steps 1 and 2,

$$\sum_F s_F |\langle \nu_F, u\rangle| = \sum_F \mathrm{vol}_{d-1}(\pi(F)) = \int_{u^\perp} \mathrm{mult}(y)\, dy = 2\,\mathrm{vol}_{d-1}(\pi(P)).$$

So $\mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} P) = \frac{1}{2}\sum_F s_F |\langle \nu_F, u\rangle|$ for every unit vector u.

Step 4 (general u and the zonotope). Let Z be the right side of (4), a compact convex set. By (S1) and (S3), $h_Z(u) = \frac{1}{2}\sum_F s_F |\langle \nu_F, u\rangle|$ for all $u \in \mathbb{R}^d$. This function is positively homogeneous of degree one, and by Step 3 it equals $\mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} P)$ on unit vectors. That is exactly the definition of the support function of ΠP: the projection onto $u^\perp$ enters only for unit u, and elsewhere $h_{\Pi P}$ is defined by homogeneity. So ΠP exists and equals Z, uniquely by (S4), and (3) holds for all $u \in \mathbb{R}^d$. □

2.3 The product identity

The next proposition is the mechanism behind both main results. It is used in the proofs of Theorem 1 and Corollary 6. The hypothesis $r, s \geq 2$ is there only because projection bodies are defined in dimension at least two.

Proposition 3. *Let $A \subset \mathbb{R}^r$ and $B \subset \mathbb{R}^s$ be full-dimensional convex polytopes, where $r, s \geq 2$. In the orthogonal product space $\mathbb{R}^{r+s} = \mathbb{R}^r \times \mathbb{R}^s$,*

$$\Pi(A \times B) = (|B|\Pi A) \times (|A|\Pi B). \tag{5}$$

In particular,

$$R_{r+s}(A \times B) = R_r(A) R_s(B). \tag{6}$$

Proof. Plan. Find the facets of $A \times B$ with their normals and areas (Steps 1–2), insert them into the facet formula (Step 3), and take volumes (Step 4). As in the source, the bare letter s is the dimension of B, while s_F, s_G (always with a subscript) are facet areas.

Step 1 (facets of $A \times B$). *The facets of $A \times B$ are the sets $F \times B$, with outward unit normal $(\nu_F, 0)$, for facets F of A, and the sets $A \times G$, with outward unit normal $(0, \nu_G)$, for facets G of B.*

For $u \in \mathbb{R}^r$ let A_u be the set of maximizers of $\langle u, \cdot\rangle$ on A, and define B_v likewise. Thus $A_0 = A$. For $u \neq 0$, A_u lies in the hyperplane $\{x : \langle u, x\rangle = h_A(u)\}$, so $\dim A_u \leq r - 1$. Let Φ be a facet of $A \times B$ with outward unit normal (u, v). By (P1), Φ is the set of maximizers of $(a, b) \mapsto \langle u, a\rangle + \langle v, b\rangle$ on $A \times B$. The two terms are maximized independently, so $\Phi = A_u \times B_v$. The affine hull of a product is the product of the affine hulls, so $\dim \Phi = \dim A_u + \dim B_v$. If $u \neq 0 \neq v$, this is at most $r + s - 2$, contradicting $\dim \Phi = r + s - 1$. So exactly one of u, v vanishes.

Suppose $v = 0$. Then $B_v = B$ and $\dim A_u = r - 1$, so A_u is a facet F of A, and $\|u\| = 1$. The hyperplane $\{x : \langle u, x\rangle = h_A(u)\}$ contains the $(r-1)$-dimensional set F, so it equals H_F, and $u = \pm\nu_F$. Both u and ν_F have A on the side where their functional is at most its value on H_F, and $A \not\subset H_F$. Hence $u = \nu_F$. The case $u = 0$ is symmetric. Conversely, $F \times B$ is the set of maximizers of $(\nu_F, 0)$ and has dimension $r + s - 1$, so it is a facet; likewise $A \times G$.

Step 2 (areas). $\mathrm{vol}_{r+s-1}(F \times B) = s_F|B|$ and $\mathrm{vol}_{r+s-1}(A \times G) = |A|s_G$. Indeed, $F \times B$ lies in $H_F \times \mathbb{R}^s$. Because the two coordinate spaces are orthogonal, an isometry $H_F \cong \mathbb{R}^{r-1}$ times the

identity of $\mathbb{R}^s$ is an isometry $H_F \times \mathbb{R}^s \cong \mathbb{R}^{r-1} \times \mathbb{R}^s$, and Lebesgue measure on the latter is the product measure. Hence the area-normal vectors of the facets of $A \times B$ are

$$(|B| s_F \nu_F, 0) \quad \text{and} \quad (0, |A| s_G \nu_G).$$

Step 3 (the projection body). By (3) for $A \times B$, then for A and for B,

$$h_{\Pi(A\times B)}(u,v) = \frac{1}{2}\sum_F |B| s_F |\langle \nu_F, u\rangle| + \frac{1}{2}\sum_G |A| s_G |\langle \nu_G, v\rangle| = |B| h_{\Pi A}(u) + |A| h_{\Pi B}(v).$$

By (S1) and (S2), the right side is the support function of $(|B|\Pi A) \times (|A|\Pi B)$ at (u, v). So (S4) gives (5).

Step 4 (volumes). Volume is multiplicative on products, and scaling by $t > 0$ multiplies j-dimensional volume by t^j. Since $\Pi A \subset \mathbb{R}^r$ and $\Pi B \subset \mathbb{R}^s$, we get $|\Pi(A \times B)| = |B|^r |\Pi A| \cdot |A|^s |\Pi B|$. Also $|A \times B| = |A||B|$. Hence

$$R_{r+s}(A \times B) = \frac{|B|^r |A|^s |\Pi A||\Pi B|}{(|A||B|)^{r+s-1}} = \frac{|\Pi A|}{|A|^{r-1}} \cdot \frac{|\Pi B|}{|B|^{s-1}},$$

because $|A|^s/|A|^{r+s-1} = |A|^{-(r-1)}$ and $|B|^r/|B|^{r+s-1} = |B|^{-(s-1)}$. □

2.4 Affine covariance

To use simplices as a benchmark we need all d-simplices to have the same value of R_d. The next lemma supplies this. Its polytope case (Steps 1–4) is used once, in the last step of Proposition 5. Its general case (Steps 5–8) is used nowhere else.

Lemma 4. *Let $P \subset \mathbb{R}^d$, $d \geq 2$, be a convex body. Then ΠP exists, and for every invertible linear map $L\colon \mathbb{R}^d \to \mathbb{R}^d$,*

$$\Pi(LP) = |\det L| L^{-t} \Pi P, \qquad R_d(LP) = R_d(P),$$

where L^{-t} denotes the inverse transpose. Translations also preserve R_d.

Proof. Plan. Translations are immediate (Step 1). For a polytope we compute how L changes the unit normal and the area of each facet (Steps 2–3). Each area-normal vector is multiplied by $M := |\det L| L^{-t}$, so (4) gives the claim (Step 4). A general body is squeezed between a polytope and a slightly larger dilate of it (Step 5). This yields the existence of ΠP (Step 6) and lets the identity pass to the limit (Steps 7–8).

Step 1 (translations). For $t \in \mathbb{R}^d$, $\mathrm{proj}_{u^\perp}(P + t)$ is a translate of $\mathrm{proj}_{u^\perp} P$. So $P + t$ and P have the same projection volumes, hence the same projection body. Also $|P + t| = |P|$, so $R_d(P + t) = R_d(P)$.

Polytope case. In Steps 2–4, P is a full-dimensional polytope. Then LP is a polytope whose facets are the images LF of the facets F of P.

Step 2 (normals). *The outward unit normal of LF is $\nu_{LF} = L^{-t}\nu_F / \|L^{-t}\nu_F\|$.* For $y = Lx$, $\langle \nu_F, x\rangle = \langle \nu_F, L^{-1}y\rangle = \langle L^{-t}\nu_F, y\rangle$. So the relations $\langle \nu_F, x\rangle = h_P(\nu_F)$ on F and $\langle \nu_F, x\rangle \leq h_P(\nu_F)$ on P become $\langle L^{-t}\nu_F, y\rangle = h_P(\nu_F)$ on LF and $\leq$ on LP. Thus $L^{-t}\nu_F$ is an outward normal of LF, and we normalize it. The argument does not involve $\det L$, so it holds for either sign of $\det L$.

Step 3 (areas). $s_{LF} = |\det L| \, \|L^{-t}\nu_F\| \, s_F$. Let $Y = F + [0, \nu_F]$ be the prism of height one over F, so $|Y| = s_F$. Then $LY = LF + [0, L\nu_F]$ and $|LY| = |\det L| s_F$. Measure height above the hyperplane

of LF by $g(y) = \langle \nu_{LF}, y\rangle - c$, where c is the constant value of $\langle \nu_{LF}, \cdot\rangle$ on LF. The set LY is the union of the translates $LF + tL\nu_F$, $t \in [0,1]$, and $g \equiv t\eta$ on the translate with parameter t, where

$$\eta = \langle L\nu_F, \nu_{LF}\rangle = \frac{\langle L\nu_F, L^{-t}\nu_F\rangle}{\|L^{-t}\nu_F\|} = \frac{\langle \nu_F, L^tL^{-t}\nu_F\rangle}{\|L^{-t}\nu_F\|} = \frac{1}{\|L^{-t}\nu_F\|} > 0.$$

Since $\eta \neq 0$, different t give different heights. So for each $h \in [0,\eta]$ the slice $\{y \in LY : g(y) = h\}$ is exactly the translate $LF + (h/\eta)L\nu_F$, of $(d-1)$-volume s_{LF}, and the slice is empty for other h. Fubini in the direction ν_{LF} gives $|LY| = \int_0^\eta s_{LF}\, dh = s_{LF}\,\eta$. Thus $s_{LF} = |\det L| s_F/\eta$, which is the claim.

Step 4 (conclusion for polytopes). By Steps 2 and 3, $s_{LF}\nu_{LF} = |\det L| L^{-t}(s_F\nu_F) = M(s_F\nu_F)$, also when $\det L < 0$. A linear map sends segments to segments and commutes with Minkowski sums, so (4) for LP and for P gives

$$\Pi(LP) = \sum_F \Big[-\tfrac{1}{2}Ms_F\nu_F, \tfrac{1}{2}Ms_F\nu_F\Big] = M\sum_F \Big[-\tfrac{1}{2}s_F\nu_F, \tfrac{1}{2}s_F\nu_F\Big] = M\,\Pi P.$$

Now $|\det M| = |\det L|^d\,|\det L|^{-1} = |\det L|^{d-1}$, so $|\Pi(LP)| = |\det L|^{d-1}|\Pi P|$. Also $|LP|^{d-1} = |\det L|^{d-1}|P|^{d-1}$. Dividing, $R_d(LP) = R_d(P)$.

General case. Now P is an arbitrary convex body. By Step 1 we may translate P so that 0 is an interior point. This changes neither side of the covariance identity, since $L(P+t) = LP + Lt$.

Step 5 (approximating polytopes). *There are $a > 0$, polytopes $P_1 \subset P_2 \subset \cdots$ and numbers $\varepsilon_m \to 0$ with*

$$aB_2^d \subset P_m \subset P \subset P_m + \varepsilon_m B_2^d \subset \mu_m P_m, \qquad \mu_m := 1 + \varepsilon_m/a \longrightarrow 1. \tag{7}$$

Construction. Choose $a > 0$ with $P_0 := [-a,a]^d \subset P$; then $aB_2^d \subset P_0$. For each $m \geq 1$, compactness gives a finite set $N_m \subset P$ such that every point of P is within $1/m$ of N_m. Put $P_m = \operatorname{conv}(P_0 \cup N_1 \cup \cdots \cup N_m)$ and $\varepsilon_m = 1/m$. Then P_m is the convex hull of finitely many points, the sequence is increasing, and $aB_2^d \subset P_m \subset P$ by convexity of P. Also $P \subset N_m + \varepsilon_m B_2^d \subset P_m + \varepsilon_m B_2^d$. Finally $\varepsilon_m B_2^d = (\varepsilon_m/a)\,aB_2^d \subset (\varepsilon_m/a)P_m$, and $P_m + (\varepsilon_m/a)P_m = \mu_m P_m$ because P_m is convex.

Fix also $\rho > 0$ with $P \subset \rho B_2^d$.

Step 6 (existence of ΠP). For unit u put $\varphi(u) = \operatorname{vol}_{d-1}(\operatorname{proj}_{u^\perp} P)$ and $\varphi_m(u) = \operatorname{vol}_{d-1}(\operatorname{proj}_{u^\perp} P_m) = h_{\Pi P_m}(u)$; here ΠP_m exists by Lemma 2. Projecting (7) gives $\operatorname{proj}_{u^\perp} P_m \subset \operatorname{proj}_{u^\perp} P \subset \mu_m \operatorname{proj}_{u^\perp} P_m$. Taking $(d-1)$-volumes,

$$\varphi_m \leq \varphi \leq \mu_m^{d-1}\varphi_m, \qquad \text{so} \qquad 0 \leq \varphi - \varphi_m \leq (\mu_m^{d-1} - 1)\kappa_{d-1}\rho^{d-1}. \tag{8}$$

The last bound uses $\varphi_m(u) \leq \operatorname{vol}_{d-1}(\operatorname{proj}_{u^\perp} \rho B_2^d) = \kappa_{d-1}\rho^{d-1}$. Thus $\varphi_m \to \varphi$ uniformly on S^{d-1}. Moreover:

- $\varphi_m \leq \varphi_{m+1}$ because $P_m \subset P_{m+1}$, so $\Pi P_m \subset \Pi P_{m+1}$ by (S4).
- $\kappa_{d-1}a^{d-1} \leq \varphi_m \leq \kappa_{d-1}\rho^{d-1}$ on S^{d-1} because $aB_2^d \subset P_m \subset \rho B_2^d$. By (S3) and (S4), $\kappa_{d-1}a^{d-1}B_2^d \subset \Pi P_m \subset \kappa_{d-1}\rho^{d-1}B_2^d$.

Let C be the closure of $\bigcup_m \Pi P_m$. An increasing union of convex sets is convex, and so is its closure. With the second bullet, C is a compact convex set with nonempty interior. The supremum of $\langle u, \cdot\rangle$ over a set equals its supremum over the closure, and the φ_m increase. So for unit u,

$h_C(u) = \sup_m h_{\Pi P_m}(u) = \lim_m \varphi_m(u) = \varphi(u)$. Thus C has the support function required of ΠP, that is, ΠP exists and equals C. By (8) and (S4),

$$\Pi P_m \subset \Pi P \subset \mu_m^{d-1}\Pi P_m, \qquad |\Pi P_m| \le |\Pi P| \le \mu_m^{d(d-1)}|\Pi P_m|, \qquad |P_m| \le |P| \le \mu_m^d |P_m|. \tag{9}$$

The last chain comes from (7).

Step 7 (covariance). Apply L to (7): $LP_m \subset LP \subset \mu_m LP_m$, with polytopes LP_m. Let $\psi(u) = \mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} LP)$ for unit u. Exactly as for (8), $h_{\Pi(LP_m)}(u) \le \psi(u) \le \mu_m^{d-1} h_{\Pi(LP_m)}(u)$, so $h_{\Pi(LP_m)}(u) \to \psi(u)$. On the other hand, Step 4 and (S2) give

$$h_{\Pi(LP_m)}(u) = h_{M\Pi P_m}(u) = h_{\Pi P_m}(M^t u) \longrightarrow h_{\Pi P}(M^t u) = h_{M\Pi P}(u).$$

The convergence holds because $h_{\Pi P_m} \to h_{\Pi P}$ on S^{d-1} by Step 6, hence on $\mathbb{R}^d$ by homogeneity. Therefore $\psi = h_{M\Pi P}$ on S^{d-1}. So $M\Pi P$ is the projection body of LP: $\Pi(LP) = |\det L| L^{-t}\Pi P$.

Step 8 (invariance of R_d). Two arguments are available; each is complete on its own.

(a) The source's argument. By (9), $\mu_m^{-d(d-1)} R_d(P_m) \le R_d(P) \le \mu_m^{d(d-1)} R_d(P_m)$, so $R_d(P_m) \to R_d(P)$. The sandwich $LP_m \subset LP \subset \mu_m LP_m$ and the inequalities for ψ in Step 7 give (9) with LP_m, LP in place of P_m, P in the same way, so $R_d(LP_m) \to R_d(LP)$. Since $R_d(LP_m) = R_d(P_m)$ by Step 4, $R_d(LP) = R_d(P)$.

(b) A direct cross-check. Step 7 gives $\Pi(LP) = M\,\Pi P$ for the convex body P. Hence $|\Pi(LP)| = |\det M|\,|\Pi P| = |\det L|^{d-1}|\Pi P|$, and $|LP|^{d-1} = |\det L|^{d-1}|P|^{d-1}$. These are the only facts used at the end of Step 4, and none of them requires P to be a polytope. □

3 The simplex value

We now compute the value to which the product will be compared. The projection body of a simplex is a coordinate cube plus one extra segment. The proposition is used in the proofs of Theorem 1 and Corollary 6, which need only the first sentence, for the standard simplex.

Proposition 5. *For $d \ge 2$, the standard simplex $T_d = \mathrm{conv}(0, e_1, \dots, e_d) \subset \mathbb{R}^d$ satisfies*

$$|T_d| = \frac{1}{d!}, \qquad |\Pi T_d| = \frac{d+1}{((d-1)!)^d}, \qquad R_d(T_d) = c_d.$$

Every full-dimensional d-simplex has the same value c_d.

Proof. Plan. Compute the $d+1$ area-normal vectors of T_d (Steps 1–2). Recognize ΠT_d via (4) as a translate of $([0,1]^d + [0,w])/(d-1)!$ with $w = e_1 + \cdots + e_d$ (Step 3). Compute the volume of $[0,1]^d + [0,w]$ by slicing along w (Steps 4–5), and assemble (Steps 6–7).

Step 1 (volume). $|T_j| = 1/j!$ *for all* $j \ge 1$. This holds for $T_1 = [0,1]$. For $j \ge 2$, the slice of T_j at height $x_j = t \in [0,1]$ is a copy of $(1-t)T_{j-1}$, so $|T_j| = \int_0^1 (1-t)^{j-1}|T_{j-1}|\,dt = |T_{j-1}|/j$. Induction gives the claim.

Step 2 (area-normal vectors). We have $T_d = \{x : x_i \ge 0\ (1 \le i \le d),\ x_1 + \cdots + x_d \le 1\}$, with $d+1$ facets.

- *Coordinate facets.* The facet in $\{x_i = 0\}$ is $\mathrm{conv}(0, e_j : j \ne i)$, a copy of T_{d-1}. Its outward unit normal is $-e_i$ and its area is $1/(d-1)!$ by Step 1. Area-normal vector: $-e_i/(d-1)!$.

- *Slanted facet.* $F_0 = \operatorname{conv}(e_1, \dots, e_d)$ lies in $\{x_1 + \cdots + x_d = 1\}$ and has outward unit normal $w/\sqrt{d}$. Dropping the last coordinate is the orthogonal projection onto $e_d^{\perp}$. It maps F_0 bijectively onto the standard simplex $T_{d-1} \subset \mathbb{R}^{d-1}$, with inverse $x' \mapsto (x', 1 - \sum_i x'_i)$. By Step 1 of Lemma 2, with $u = e_d$, this projection multiplies areas on the hyperplane of F_0 by $|\langle w/\sqrt{d}, e_d\rangle| = 1/\sqrt{d}$. Hence $1/(d-1)! = s_{F_0}/\sqrt{d}$, that is, $s_{F_0} = \sqrt{d}/(d-1)!$. Area-normal vector: $\dfrac{\sqrt{d}}{(d-1)!} \cdot \dfrac{w}{\sqrt{d}} = \dfrac{w}{(d-1)!}$.

Sanity check: the $d+1$ area-normal vectors sum to $(-e_1 - \cdots - e_d + w)/(d-1)! = 0$, as they must for a closed polytope.

Step 3 (ΠT_d as a zonotope). Put $\theta = 1/(d-1)!$. By (4),

$$\Pi T_d = \sum_{i=1}^{d} \Big[-\tfrac{\theta}{2} e_i, \tfrac{\theta}{2} e_i\Big] + \Big[-\tfrac{\theta}{2} w, \tfrac{\theta}{2} w\Big].$$

The sign of the generator $-e_i$ is irrelevant, because a centered segment is unchanged when its generator is negated. Since $[-\frac{1}{2}v, \frac{1}{2}v] = -\frac{1}{2}v + [0, v]$ and $\sum_i [0, e_i] = [0,1]^d$, the body ΠT_d is a translate of

$$\frac{[0,1]^d + [0, w]}{(d-1)!}.$$

Step 4 (extrusion volume). Let $Q = [0,1]^d$. *Then* $|Q + [0,w]| = 1 + h_{\Pi Q}(w)$. Let $\hat{w} = w/\|w\|$ and $\pi = \operatorname{proj}_{w^\perp}$. For a set S and $y \in w^\perp$, let $S_y = \{t \in \mathbb{R} : y + t\hat{w} \in S\}$ be its fiber over y. Note that $[0, w] = \{\sigma \hat{w} : 0 \le \sigma \le \|w\|\}$.

- If $y \in \pi(Q)$, then Q_y is a nonempty compact interval $[t_0, t_1]$, possibly a single point, because Q is compact and convex. A point $y + t\hat{w}$ lies in $Q + [0,w]$ if and only if $y + t\hat{w} = x + \sigma\hat{w}$ with $x \in Q$ and $\sigma \in [0, \|w\|]$, that is, if and only if $t - \sigma \in [t_0, t_1]$ for some $\sigma \in [0, \|w\|]$. So $(Q + [0,w])_y = [t_0, t_1 + \|w\|]$, of length $(t_1 - t_0) + \|w\|$.
- If $y \notin \pi(Q)$, then $(Q + [0,w])_y = \emptyset$, because $\pi(w) = 0$ gives $\pi(Q + [0,w]) = \pi(Q)$.

By Fubini, applied to $Q + [0,w]$ and to Q,

$$|Q + [0,w]| = \int_{\pi(Q)} \big(\operatorname{length}(Q_y) + \|w\|\big)\, dy = |Q| + \|w\| \operatorname{vol}_{d-1}(\operatorname{proj}_{w^\perp} Q) = 1 + h_{\Pi Q}(w).$$

The last equality uses $|Q| = 1$, the definition of ΠQ at the unit vector $\hat{w}$, and homogeneity: $\|w\| h_{\Pi Q}(\hat{w}) = h_{\Pi Q}(w)$.

Step 5 ($h_{\Pi Q}(w) = d$). The cube has $2d$ facets, with unit normals $\pm e_i$ and area one, and $|\langle \pm e_i, w\rangle| = 1$. By (3), $h_{\Pi Q}(w) = \frac{1}{2} \cdot 2d \cdot 1 = d$. Hence $|Q + [0,w]| = d + 1$. Figure 1 shows the case $d = 2$.

Step 6 (assembling). Volume is translation invariant, and dilation by $1/(d-1)!$ in $\mathbb{R}^d$ multiplies volume by $((d-1)!)^{-d}$. So Steps 3–5 give $|\Pi T_d| = (d+1)/((d-1)!)^d$. With $|T_d| = 1/d!$ and $(d-1)! = d!/d$,

$$R_d(T_d) = \frac{(d+1)(d!)^{d-1}}{((d-1)!)^d} = \frac{(d+1)(d!)^{d-1} d^d}{(d!)^d} = \frac{(d+1)d^d}{d!} = c_d.$$

Sanity check for $d = 2$: T_2 has area $\frac{1}{2}$, ΠT_2 is a translate of the hexagon of Figure 1 with area 3, and $3/\frac{1}{2} = 6 = c_2$.

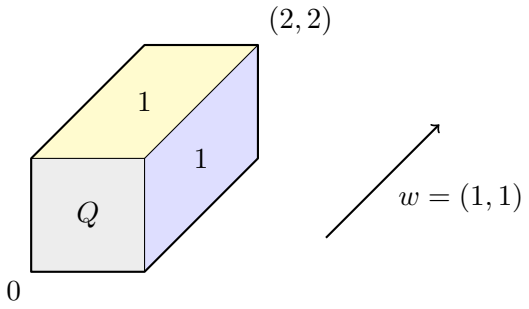


Figure 1: The planar extrusion $Q + [0, (1, 1)]$, with $Q = [0, 1]^2$, consists of the unit square and two parallelograms of area one. Its area is 3. The fiber argument in Proposition 5 proves the corresponding volume $d + 1$ in every dimension.

Step 7 (all simplices). A full-dimensional simplex $\operatorname{conv}(v_0, \dots, v_d)$ equals $v_0 + LT_d$, where L is the linear map with $Le_i = v_i - v_0$. This map is invertible because the v_i are affinely independent. By Lemma 4 (polytope case, Steps 1 and 4), $R_d(v_0 + LT_d) = R_d(T_d) = c_d$. □

4 The counterexample and exponential excess

The geometric work is complete. What remains is arithmetic with the constants c_d.

Proof of Theorem 1. **Step 1.** T_{10} is a full-dimensional convex polytope in $\mathbb{R}^{10}$, and $10 \geq 2$. So $K = T_{10} \times T_{10}$ is a convex body in $\mathbb{R}^{20}$, and Proposition 3 with $A = B = T_{10}$, followed by Proposition 5, gives

$$R_{20}(K) = R_{10}(T_{10})^2 = c_{10}^2 = \left(\frac{11 \cdot 10^{10}}{10!} \right)^2 .$$

Step 2. Dividing by $c_{20} = 21 \cdot 20^{20}/20!$ and using $10^{20}/20^{20} = 2^{-20}$ and $20!/(10!)^2 = \binom{20}{10}$,

$$\frac{R_{20}(K)}{c_{20}} = \frac{121 \cdot 10^{20} \cdot 20!}{(10!)^2 \cdot 21 \cdot 20^{20}} = \frac{121 \binom{20}{10}}{21 \cdot 2^{20}}.$$

Step 3. $\binom{20}{10} = 184\,756$ and $2^{20} = 1\,048\,576$. Then $121 \cdot 184\,756 = 22\,355\,476$ and $21 \cdot 1\,048\,576 = 22\,020\,096$, so

$$121 \cdot 184\,756 - 21 \cdot 1\,048\,576 = 335\,380 > 0.$$

The ratio is therefore $22\,355\,476/22\,020\,096 > 1$ (about 1.015). That is, $|\Pi K| > c_{20}|K|^{19}$, a strict violation of (2). □

The general product test. For $r, s \geq 2$, the same two propositions give $R_{r+s}(T_r \times T_s) = c_r c_s$. Dividing by c_{r+s} and using $(r + s)!/(r!\, s!) = \binom{r+s}{r}$,

$$\frac{R_{r+s}(T_r \times T_s)}{c_{r+s}} = \frac{c_r c_s}{c_{r+s}} = \frac{(r+1)(s+1)}{r+s+1} \binom{r+s}{r} \frac{r^r s^s}{(r+s)^{r+s}}.$$

Since $T_r \times T_s$ is a convex body in $\mathbb{R}^{r+s}$, any universal upper constant in dimension $r + s$ must be at least $c_r c_s$. The case $r = s = 10$ is Theorem 1.

Iterating the product identity gives the following quantitative consequence.

Corollary 6. *There exist a constant $\lambda > 1$ and an integer $N \geq 20$ such that, for every integer $n \geq N$, some full-dimensional convex polytope $K_n \subset \mathbb{R}^n$ satisfies*

$$\frac{R_n(K_n)}{R_n(T_n)} \geq \lambda^n.$$

The polytopes K_n may be chosen as Cartesian products of simplices.

Proof. Plan. Take k factors T_{10} and one more simplex of dimension between 10 and 19 to reach dimension n (Step 1). Bound c_n above by $(n+1)\beta^n$ with $\beta = 11/4$ (Step 2). Check that $c_{10} > \beta^{10}$, so that each ten-dimensional factor contributes more than β^{10} (Step 3). Then compare (Steps 4–5).

Step 1 (the bodies). For $r \in \{0, \dots, 9\}$ put $d_r = 10 + r$. Every integer $n \geq 20$ has a unique representation $n = 10k + d_r$ with $k \geq 1$ and $r \in \{0, \dots, 9\}$: divide $n - 10 \geq 10$ by 10 with remainder r. Let

$$K_n = T_{10}^k \times T_{d_r} \subset \mathbb{R}^n,$$

where T_{10}^k is the k-fold Cartesian product. A product of full-dimensional convex polytopes is one, and all factor dimensions are at least $10 \geq 2$. So Proposition 3 applies k times, and with Proposition 5,

$$R_n(K_n) = c_{10}^k \, c_{d_r}, \qquad R_n(T_n) = c_n.$$

Step 2 (upper bound for c_n). Put $\beta = 11/4$. *Then $e < \beta$ and $c_n < (n+1)\beta^n$ for $n \geq 2$.* For $i \geq 0$ we have $(3+i)! = 6 \cdot \prod_{m=4}^{3+i} m \geq 6 \cdot 4^i$: the product has i factors, each at least 4, and is empty for $i = 0$. Hence

$$\frac{1}{(3+i)!} \leq \frac{1}{6} \, 4^{-i} \leq \frac{1}{6} \, 3^{-i},$$

and the second inequality is strict for $i \geq 1$. (This is the source's remark that successive ratios in the tail are less than 1/3.) Summing over $i \geq 0$,

$$e = \frac{5}{2} + \sum_{j=3}^{\infty} \frac{1}{j!} < \frac{5}{2} + \frac{1}{6} \sum_{i=0}^{\infty} 3^{-i} = \frac{5}{2} + \frac{1}{6} \cdot \frac{3}{2} = \frac{11}{4} = \beta.$$

Next, $n^n/n!$ is one term of the series $e^n = \sum_{j \geq 0} n^j/j!$, whose terms are all positive. So $n^n/n! < e^n$ and

$$c_n = (n+1)\frac{n^n}{n!} < (n+1)e^n < (n+1)\beta^n.$$

Step 3 (the ten-dimensional block). We have $10! = 3\,628\,800$, so $10^{10}/10! = 10^8/36\,288$. With $36\,288 = 2^6 \cdot 567$ and $10^8 = 2^8 \cdot 5^8$, this is $4 \cdot 5^8/567 = 1\,562\,500/567$. Hence

$$c_{10} = \frac{11 \cdot 10^{10}}{10!} = \frac{17\,187\,500}{567}, \qquad \alpha := \frac{c_{10}}{\beta^{10}} = \frac{2^2 \cdot 5^8 \cdot 11 \cdot 4^{10}}{567 \cdot 11^{10}} = \frac{2^{22} \cdot 5^8}{567 \cdot 11^9} = \frac{1\,638\,400\,000\,000}{1\,336\,956\,340\,797} > 1.$$

Here $2^{22} \cdot 5^8 = 2^{14} \cdot 10^8$ and $2^{14} = 16\,384$. Also $11^9 = 2\,357\,947\,691$ and $567 \cdot 2\,357\,947\,691 = 1\,336\,956\,340\,797$. So $\alpha \approx 1.2255$.

Step 4 (comparison). Fix λ with $1 < \lambda < \alpha^{1/10}$, so that $\alpha/\lambda^{10} > 1$. For $n = 10k + d_r$, Steps 1 and 2 give

$$\frac{R_n(K_n)}{R_n(T_n)\lambda^n} = \frac{c_{10}^k \, c_{d_r}}{c_n \lambda^n} > \frac{c_{10}^k \, c_{d_r}}{(n+1)\beta^n \lambda^n}.$$

Substitute $c_{10}^k = \alpha^k \beta^{10k}$, $\beta^n = \beta^{10k}\beta^{d_r}$, $\lambda^n = \lambda^{10k}\lambda^{d_r}$ and $n+1 = 10k + d_r + 1$. The factor β^{10k} cancels, and

$$\frac{R_n(K_n)}{R_n(T_n)\lambda^n} > c_{d_r}\beta^{-d_r}\lambda^{-d_r}\,\frac{(\alpha/\lambda^{10})^k}{10k + d_r + 1}.$$

Step 5 (thresholds). Fix r. The right side is a positive constant times $\gamma^k/(10k + d_r + 1)$ with $\gamma = \alpha/\lambda^{10} > 1$. It tends to infinity as $k \to \infty$, since an exponential with base > 1 outgrows a linear function. So there is $k_r \geq 1$ such that the right side is at least 1 for all $k \geq k_r$. Let $N = \max_{0\leq r\leq 9}(10k_r + d_r)$; then $N \geq 20$. If $n \geq N$ and $n = 10k + d_r$, then $10k + d_r \geq 10k_r + d_r$, so $k \geq k_r$. Step 4 then gives $R_n(K_n)/R_n(T_n) > \lambda^n$. □

Verification checklist

No step of this rewrite is marked as unverified. A sceptical reader should check the following load-bearing points first.

1. **The factor $\frac{1}{2}$ in the facet formula.** Almost every line parallel to u through P meets exactly two facets, so $\sum_F s_F|\langle \nu_F, u\rangle| = 2\,\mathrm{vol}_{d-1}(\mathrm{proj}_{u^\perp} P)$. See Lemma 2, Steps 1–3.
2. **Facets of a product and the weights in (5).** The facets of $A \times B$ are exactly $F \times B$ and $A \times G$, with areas $s_F|B|$ and $|A|s_G$. The exponent bookkeeping $|B|^r|A|^s/(|A||B|)^{r+s-1} = |A|^{-(r-1)}|B|^{-(s-1)}$ must come out. See Proposition 3, Steps 1, 2 and 4.
3. **The slanted facet of the simplex.** Its area is $\sqrt{d}/(d-1)!$, so its area-normal vector is $w/(d-1)!$, and the $d+1$ area-normal vectors sum to zero. See Proposition 5, Step 2.
4. **The extrusion volume $|[0,1]^d + [0,w]| = d+1$.** It rests on the fiber-length identity and $h_{\Pi Q}(w) = d$. See Proposition 5, Steps 4–5; test case $d = 2$ in Figure 1 and Step 6.
5. **The arithmetic of Theorem 1.** $c_{10}^2/c_{20} = 121\binom{20}{10}/(21\cdot 2^{20})$ and $121\cdot 184\,756 - 21\cdot 1\,048\,576 = 335\,380 > 0$. See the proof of Theorem 1, Steps 2–3.
6. **Dependence.** Theorem 1 needs only items 1–5. Lemma 4 enters only through "every simplex has value c_d" (Proposition 5, Step 7), and there only in its polytope case. Its key computation is the prism height $\langle L\nu_F, \nu_{LF}\rangle = 1/\|L^{-t}\nu_F\|$ in Step 3 of Lemma 4.
7. **For Corollary 6.** Check $e < 11/4$ and $c_n < (n+1)(11/4)^n$ (Step 2), the constant $\alpha = 2^{22}5^8/(567\cdot 11^9) > 1$ (Step 3), and the cancellation of β^{10k} (Step 4).

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